Xing 2021 lung cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 18 samples (≥100 malignant cells) intermix per-neighbourhood on one continuous manifold: iLISI 1.4→4.5 across the θ 0→10 sweep, diff-sample kNN fraction 0.72 (of a 0.91 ceiling), a single connected kNN component, 0 detached islands. Cell cycle (~4.5%) is diffuse and does not fragment the cloud. Read on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

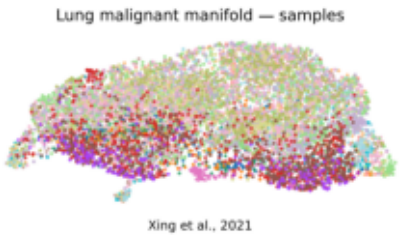
The embryonic axis is present: a Gast/pSom-high origin pole giving way to an inverse terminal Neural-crest gradient (Crest ρ vs pseudotime +0.69, Crest-high decile at pseudotime 0.92; joint pSom&Crest-high depleted, enrichment 0.28). Gast and pSom cross-repress (smoothed r=−0.52, −0.78 at the origin) but do NOT resolve into two competing arms — a branching-capable PPT fit gives no stable fork (forks grow 2→5 with node density on one roundish mass). Non-branching case, GO; no inversion flag.

Gate 3 — cross-repression → NA

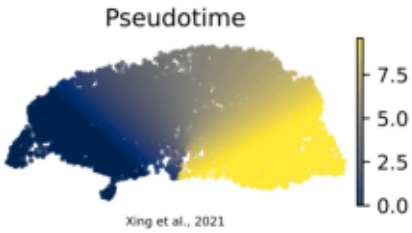
Not applicable: the manifold is non-branching (single continuous developmental gradient, no stable bifurcation), so there is no hub convergence population to test for competitive Gast↔pSom cross-repression. The branching-only correlation and modules notebooks are therefore not produced.

STEP FIGURES

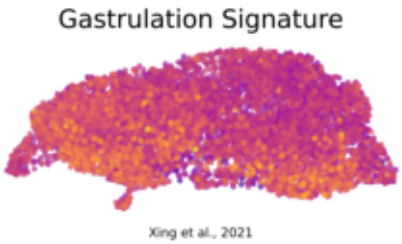
Manifold — samples



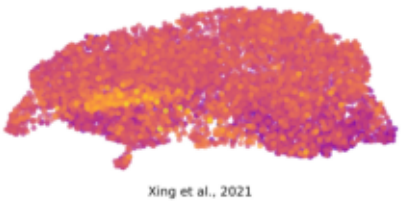
Pseudotime



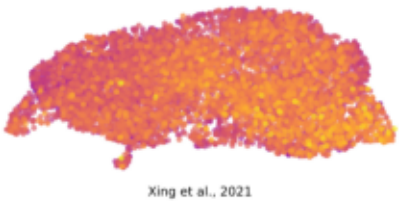
Gastrulation



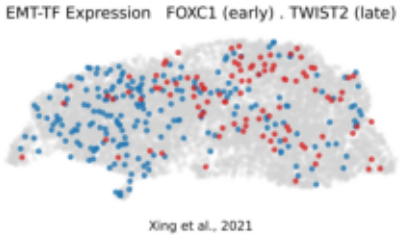
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Lung cancer (subsolid nodules + adjacent margins), 10x UMI counts — 9,268 malignant cells across 25 samples → 8,920 cells / 18 samples after the ≥100-cell floor and island removal.

Verdict.
The embryonic pattern REPRODUCES in the non-branching form: a single continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high developmental origin, with organised Gast↔pSom cross-repression that does not resolve into competing arms. EMT-TFs FOXC1 (root) → TWIST2 (terminus). Global module strength pSom>Gast~Crest (Linear-pSom/Crest).

Parameter choices

Malignant load	9,268 cells / 25 samples (publication calls)	**After thresh + islands**	8,920 cells / 18 samples
Integration	Harmony θ=10 (sweep-derived plateau)	**Embedding**	n_pcs=20, n_neighbors=30, UMAP spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Root**	curve tip: Gast/pSom-high, Crest-low
Topology	non-branching (2 tips, 0 forks, self-approach 2.7)	**EMT-TFs**	FOXC1 (early, 9.6%) · TWIST2 (late, 4.5%)